

Product Requirements Document (PRD)

Fam — Family Tree & Relationship Mapping Platform

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1. Executive Summary

1.1 Product Name

Fam

1.2 Product Vision

Fam is a simple and minimalistic family tree and relationship mapping platform designed specifically for family gatherings and events.

The product is not intended to be a full-scale event management platform.

Its purpose is to help families:

- Create family trees during gatherings
- Discover and understand relationships
- Preserve family connections digitally
- Visualize multi-generational structures
- Build shared family relationship maps

Fam focuses entirely on relationships, identity, and family structure.

The product intentionally avoids operational event-management complexity.

2. Background & Context

Most existing family-event applications focus heavily on:

- Invitations
- RSVPs
- Vendor management
- Scheduling
- Media sharing
- Logistics

However, families primarily struggle with:

- Remembering relationships
- Understanding extended family connections
- Preserving generational lineage
- Explaining family structures to younger generations
- Reconnecting with distant relatives

Fam solves this by providing a lightweight and easy-to-use family relationship mapping system.

The product is designed around simplicity, clarity, and emotional connection.

3. Product Objectives

3.1 Primary Objectives

Relationship Preservation

Digitally preserve family structures and relationships.

Family Discovery

Help users discover and understand how they are connected.

Multi-Generational Mapping

Support large and culturally complex family structures.

Simplicity

Keep the experience minimal, intuitive, and non-overwhelming.

Mobile-First Accessibility

Ensure the platform works smoothly during live family gatherings and events.

4. Product Philosophy

Fam is intentionally designed as a focused and lightweight product.

The platform should:

- Be simple
- Be fast
- Be emotionally warm
- Avoid feature overload
- Prioritize relationships over operations
- Remain usable for elderly family members
- Work well on mobile devices

Fam should never evolve into a bloated event-management suite.

The simplicity of the product is a core strategic advantage.

5. Problem Statement

Families often face the following problems:

Problem	Impact
Complex family relationships	Relationship confusion

Lack of centralized family structure	Loss of family continuity
Difficulty remembering extended relatives	Weak long-term connections
Fragmented family information	Poor relationship visibility
Existing genealogy tools are too complicated	Low adoption
Western-centric family-tree systems	Poor compatibility with Indian families

Fam addresses these challenges through a simple relationship-first platform.

6. Product Scope

6.1 In Scope

Core Modules

- Family Tree Creation
 - Relationship Mapping
 - Shared Family Trees
 - User Profiles
 - Profile Pictures
 - QR-Based Tree Joining
 - Relationship Discovery
 - Role-Based Permissions
 - Lightweight Authentication
 - Mobile-Friendly Tree Navigation
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6.2 Out of Scope

The following are intentionally excluded from the product direction:

- Full event management systems
- Vendor management
- Ticketing systems
- Scheduling workflows
- Venue management
- Large media galleries
- Video uploads
- Document uploads
- File storage systems
- Live streaming
- Enterprise workflow systems
- Social media feed systems

Fam is intentionally focused only on relationships and family-tree creation.

7. Target Audience

Primary Users

Families

Families creating and preserving relationship structures.

Multi-Generational Households

Families with large and extended structures.

Community Groups

Cultural and community groups preserving lineage and social relationships.

Secondary Users

- Elder family members
- Family historians

- Community leaders
 - Heritage researchers
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8. Product Positioning

Fam is positioned as:

“A minimalistic family relationship and family-tree platform.”

Unlike generic genealogy software, Fam focuses on:

- Simplicity
 - Accessibility
 - Indian family structures
 - Emotional usability
 - Lightweight onboarding
 - Mobile-first relationship mapping
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9. Core Product Modules

9.1 Authentication System

Features

- Email OTP Login
 - Lightweight onboarding
 - Session management
 - Role-based access
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Functional Requirements

ID	Requirement
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AUTH-1	Users must support email-based login
AUTH-2	OTP onboarding should be supported for email authentication
AUTH-3	Sessions should remain secure
AUTH-4	Access permissions should be role-based

9.2 Family Tree System

Overview

The Family Tree module is the platform’s core feature.

The system must support:

- Multi-generational structures
- Maternal and paternal relations
- Step-relations
- Joint families
- Extended relatives
- Relationship discovery
- Indian family structures

Relationship Types

Category	Examples
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Direct Relations	Father, Mother, Son
Extended Relations	Uncle, Aunt, Cousin
Marital Relations	Spouse, In-laws
Step Relations	Stepfather, Stepmother
Community Relations	Mentor, Guardian

Functional Requirements

ID	Requirement
TREE-1	Users should create family nodes
TREE-2	Relationship types must support flexibility
TREE-3	Trees should support visual rendering
TREE-4	Visibility permissions must be configurable

TREE-5	Large family trees should load efficiently
TREE-6	QR-based joining should be supported

9.3 User Profiles

Supported Profile Data

- Name
 - Profile Picture
 - Gender
 - Relationship labels
 - Birth year (optional)
 - Deceased status
 - Family branch
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Media Policy

Fam intentionally supports only profile pictures.

The platform will not support:

- Video uploads
- Document uploads
- Media galleries
- Large file storage
- Cloud-drive style features

This decision helps:

- Keep infrastructure lightweight
- Improve performance
- Reduce storage costs
- Simplify the user experience
- Keep the product relationship-focused

9.4 QR Joining System

Use Cases

- Join a family tree during an event
 - Quickly add relatives
 - Invite members into shared trees
 - Simplify onboarding during gatherings
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Functional Requirements

ID	Requirement
QR-1	Trees should generate shareable QR codes
QR-2	QR joining should work on mobile devices
QR-3	Users should join trees with minimal onboarding
QR-4	QR permissions should remain secure

9.5 Relationship Discovery

Features

- Shared family visibility

- Connection discovery
 - Multi-generation navigation
 - Relationship tracing
 - Family branch exploration
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10. Role-Based Access Control (RBAC)

Admin Role

Permissions

- Moderate platform activity
 - Manage users
 - Handle abuse reports
 - Access analytics
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Tree Organizer Role

Permissions

- Create family trees
 - Invite relatives
 - Edit relationship structures
 - Manage visibility settings
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Member Role

Permissions

- Join trees
 - View relationships
 - Update profile details
 - Add permitted connections
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11. Non-Functional Requirements

11.1 Performance

Metric	Target
App Load Time	< 3 seconds
Tree Render Time	< 2 seconds
API Response Time	< 500ms
Mobile Responsiveness	Optimized

11.2 Scalability

System should support:

- Millions of users
- Large family graphs
- Multi-generational structures
- Real-time tree updates

11.3 Security

Requirements

- Encrypted authentication
- Secure profile image storage
- Role-based access control
- Privacy-first relationship permissions

- Secure APIs
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12. Technical Architecture

Deployment Infrastructure

Hosting Stack

Fam shall be deployed using:

- Vercel for frontend hosting
- Railway for backend services and infrastructure

The architecture should remain lightweight, scalable, and optimized for rapid deployment.

Frontend

- React / Next.js
 - Tailwind CSS
 - Mobile-first architecture
 - PWA support
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Backend

- Node.js / NestJS
 - REST APIs
 - Graph-ready architecture
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Database

- PostgreSQL
- Neo4j (recommended)
- Redis caching

Cloud Infrastructure

- AWS / GCP
 - Lightweight object storage for profile pictures only
 - CDN optimization
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13. Data Model Overview

Core Entities

- User
 - FamilyNode
 - Relationship
 - FamilyTree
 - QRInvite
 - ProfileImage
 - Notification
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14. User Flow Examples

Family Gathering Flow

1. Create family tree
 2. Generate QR code
 3. Share QR during gathering
 4. Family members join tree
 5. Add relationships
 6. Explore family connections
 7. Preserve family structure digitally
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15. Monetization Strategy

Potential Pricing

Plan	Pricing
Family Tree Subscription	₹2999/event (₹1999/event initially for first 50 customers)
World Tree	₹499/month

Future Revenue Streams

- Premium relationship visualization
- Advanced family analytics
- Custom themes
- Extended tree capabilities

16. Privacy & Compliance

Privacy Principles

Fam must remain privacy-first.

Users should control:

- Tree visibility
 - Relationship visibility
 - Profile access
 - Personal information sharing
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17. Risks & Challenges

Risk	Impact	Mitigation
Complex family structures	High	Graph database design
Privacy concerns	High	Granular permissions
Elder-user usability	Medium	Minimal UI
Feature overload	High	Strict product focus

18. Success Metrics

User Metrics

- Trees created
- Active family members
- Relationship connections added
- Monthly active users
- Tree completion rate

Technical Metrics

- Tree render performance
- App stability
- Mobile responsiveness
- Infrastructure efficiency

19. MVP Scope

Must Have

- Authentication
- Family tree creation
- Relationship mapping
- QR joining
- Profile pictures
- Mobile support
- Permission controls

Nice to Have

- Multi-language support
- Advanced relationship discovery
- Tree themes
- Branch grouping

20. Future Roadmap

Phase 2

- Native mobile apps
- Smarter relationship discovery
- Better tree visualization
- Multi-language support

Phase 3

- Global relationship graph
- Heritage preservation tools

- Advanced genealogy systems
 - Family lineage insights
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21. UX Principles

Design Philosophy

Fam must feel:

- Lightweight
- Calm
- Warm
- Family-friendly
- Easy to use
- Non-technical
- Minimalistic

The interface should prioritize clarity over feature density.

UX Priorities

- Mobile-first navigation
 - Elder-friendly readability
 - Minimal onboarding friction
 - Simple relationship flows
 - Fast tree interaction
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22. Recommended Tech Stack

Layer	Recommendation
Frontend	Next.js

Backend	NodeJS
Database	PostgreSQL + Neo4j
Cache	Redis
Storage	Lightweight object storage
Infra	AWS ECS/Kubernetes

23. API Requirements

Core APIs

- Authentication API
 - Family Tree API
 - Relationship API
 - QR Join API
 - Profile API
 - Notification API
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24. Security Requirements

Security Controls

- JWT authentication
- Rate limiting
- Encryption standards

- Role isolation
 - Secure uploads for profile pictures
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25. Accessibility Requirements

Accessibility Standards

- WCAG compliance
 - Large readable typography
 - Mobile accessibility
 - Screen-reader support
 - High contrast mode
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26. Launch Strategy

Initial Target Audience

- Indian families
 - Weddings and family gatherings
 - Community groups
 - Multi-generational households
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Acquisition Channels

- Referral-based growth
 - Community associations
 - Family networks
 - Word of mouth
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27. Competitive Advantage

Key Differentiators

Feature	Fam	Traditional Event Apps
Family Tree Focus	Yes	No
Minimalistic UX	Yes	Rarely
Relationship Discovery	Yes	No
Indian Family Support	Yes	Limited
Lightweight Experience	Yes	No
Relationship-First Design	Yes	No

28. Final Product Philosophy

Fam is not an event-management platform.

It is a lightweight relationship and family-tree platform.

The product exists to help families:

- Understand relationships
- Preserve generational structures
- Discover connections
- Stay connected across generations

while remaining simple, scalable, and emotionally meaningful.

Appendix A — Suggested Initial Database Tables

- users
 - family_nodes
 - relationships
 - family_trees
 - qr_invites
 - profile_images
 - notifications
 - audit_logs
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Appendix B — Suggested Integrations

Integration	Use Case
Email OTP Authentication	User Login
Resend	OTP
Railway	Profile Pictures

Appendix C — Strategic Product Direction

Fam should evolve into:

1. Family Relationship Platform
2. Family Tree Infrastructure
3. Multi-Generational Relationship System
4. Community Connection Layer
5. Digital Family Continuity Platform